

定理 1.1 (隐函数存在定理).

设 $D = [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b]$,

$F(x, y)$ 是 D 上的函数, 满足

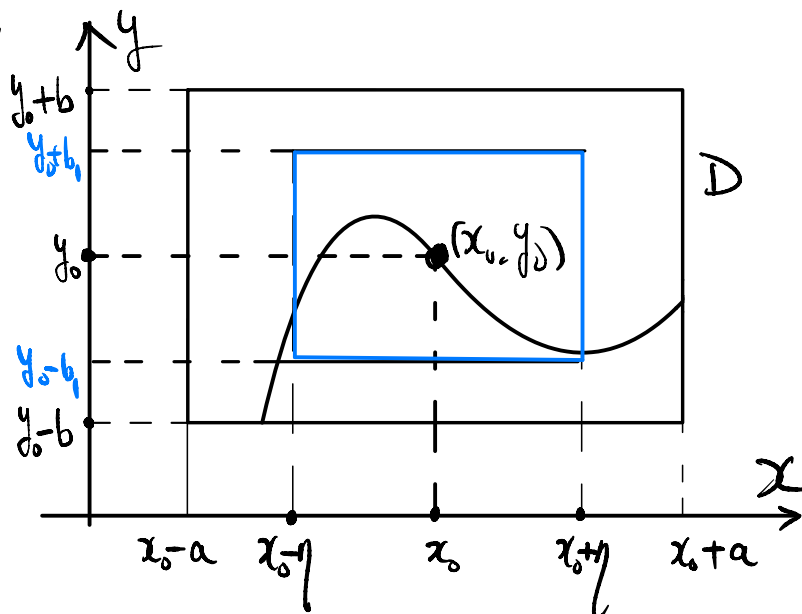
(1) $F(x, y) \in C^1(D)$. 即

F_x, F_y 在 D 中连续.

(2) $F(x_0, y_0) = 0$ 且 $(x_0, y_0) \in D$.

即 $F(x_0, y_0) = 0$,

(3) $F_y(x_0, y_0) \neq 0$.



则 $\exists \eta \in (0, a]$: $\Rightarrow F(x, y) = 0$ 确定了唯一隐函数

$y = f(x)$, $|x - x_0| \leq \eta$. 满足下列性质:

(i) $y = f(x) \in C^1[x_0 - \eta, x_0 + \eta]$. 且

$y_0 = f(x_0)$; $(x, f(x)) \in D$; $\forall x \in [x_0 - \eta, x_0 + \eta]$.

(ii) $F(x, f(x)) = 0$. $\forall x \in [x_0 - \eta, x_0 + \eta]$.

(iii) $\frac{df(x)}{dx} = -\frac{F_x}{F_y}$. $|x - x_0| \leq \eta$

证明: $y = f(x)$ 是隐函数 $\Leftrightarrow F(x, f(x)) \equiv 0$,

$\Leftrightarrow f(x) - Af(x, f(x)) \equiv f(x)$ $A \neq 0$

Step 1: 映射: $v(x) \mapsto u(x) = Tv(x) \triangleq v(x) - AF(x, v(x))$.

T 不动点 $\sum f(x)$: $\mathcal{D}_f$

$$\underline{Tf(x)} = f(x) \quad \forall |x-x_0| \leq \eta$$

$$\Leftrightarrow f(x) - AF(x, f(x)) \equiv f(x):$$

$$\Leftrightarrow f(x) \equiv 0$$

Picard 迭代格式:

$$y_0(x) \equiv y_0$$

$$y_1(x) = Ty_0(x) = y_0 - AF(x, y_0)$$

$\forall n \geq 1$:

$$y_{n+1}(x) = Ty_n(x) \triangleq y_n(x) - AF(x, y_n(x)).$$

Step 2: 估计 $y_n(x)$:

$$\exists \eta > 0, \exists k \in (0, 1): \quad \forall n \geq 1$$

$$|y_{n+1}(x) - y_n(x)| \leq k |y_n(x) - y_{n-1}(x)|, \quad \forall |x-x_0| \leq \eta$$

$$\begin{aligned} y_{n+1}(x) - y_n(x) &= Ty_n(x) - Ty_{n-1}(x) \\ &= (\underline{y_n(x)} - AF(x, y_n(x))) - (\underline{y_{n-1}(x)} - AF(x, y_{n-1}(x))) \end{aligned}$$

$$= \underbrace{(y_{n+1}(x) - y_n(x))}_{\substack{\text{F(x,y) 关于 y 可导} \\ \text{F(x,y) 关于 y 可导}}} - A \underbrace{(F(x, y_n(x)) - F(x, y_{n-1}(x)))}_{\substack{\uparrow \\ AF_y(x, \xi_n(x)) (y_n(x) - y_{n-1}(x))}}$$

$F(x, y)$ 关于 y 可导.

$$= \underline{\underline{[1 - AF_y(x, \xi_n(x))]} (y_n(x) - y_{n-1}(x))}$$

$\xi_n(x)$ 介于 $y_{n-1}(x)$ 与 $y_n(x)$ 之间.

能否取 η, b_1 使得 $\forall n \geq 1, \quad |x - x_0| < \eta, \quad |y - y_0| \leq b_1$

$|1 - AF_y(x, \xi_n(x))| \leq \frac{1}{2}$

从而 $\Rightarrow |y_{n+1}(x) - y_n(x)| \leq \frac{1}{2} |y_n(x) - y_{n-1}(x)|$ $k = \frac{1}{2}$:

这要求: $(x, y_n(x)) \in D_1 \rightarrow$ 保证迭代能进行下去.

$(x, y_n(x)) \notin D_1 \subset D \quad |1 - AF_y(x, \xi_n(x))| \leq \frac{1}{2}$

这要求 $(x, y_n(x)) \in D_1 \Leftrightarrow |x - x_0| \leq \eta_1, \quad |y_n(x) - y_0| \leq b_1$

因为 $F_y(x_0, y_0) \neq 0$:

可取 $A = (F_y(x_0, y_0))^{-1} = \frac{1}{F_y(x_0, y_0)}$

则 $1 - AF_y(x_0, y_0) = 0$.

又因为 $1 - AF_y(x, y)$ 关于 $(x, y) \in D$ 连续.

$$(x, y) \rightarrow (x_0, y_0): \underbrace{1 - AF_y(x, y) \rightarrow 1 - AF_y(x_0, y_0) = 0}$$

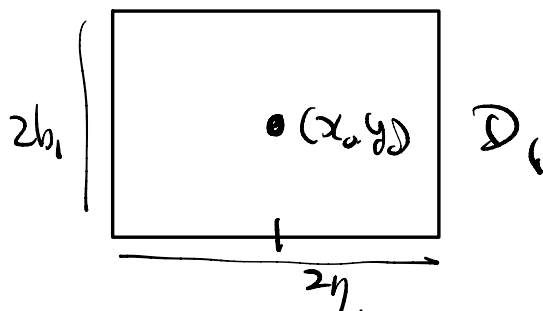
$$\therefore \exists (x_0, y_0) \in \text{令 } D_1: \begin{cases} |x - x_0| \leq \eta_1 (\leq a) \\ |y - y_0| \leq b_1 (\leq b) \end{cases}$$

$$\underbrace{|1 - AF_y(x, y)| \leq \frac{1}{2}} \text{ 选择 } \eta_1, b_1 \text{ 确定!}$$

$$\text{对于 } \exists 0 < \eta \leq \eta_1$$

$$|y_n(x) - y_0| \leq b_1$$

$$\forall (x - x_0) \leq \eta$$



$$\Rightarrow (x, y_n(x)) \in D_2 = \begin{cases} |x - x_0| \leq \eta \\ |y - y_0| \leq b_1 \end{cases}$$

$$\xi_n(x): \xi_n(x) \text{ 介于 } y_{n-1}(x) \text{ 与 } y_n(x) \text{ 之间}$$

$$\Rightarrow |\xi_n(x) - y_0| \leq b_1$$

$$\Rightarrow (x, \xi_n(x)) \in D_1 \Rightarrow |1 - AF_y(x, \xi_n(x))| \leq \frac{1}{2}$$

$$\Rightarrow |y_{n+1}(x) - y_n(x)| \leq \frac{1}{2} |y_n(x) - y_{n-1}(x)|, \forall (x - x_0) \leq \eta$$

$$\text{故 } \exists \eta \leq \eta_1: \begin{cases} |y_n(x) - y_0| \leq b_1 \\ \forall (x - x_0) \leq \eta \end{cases} \quad \forall n \geq 1$$

$$y_0(x) \equiv y_0 \text{ 不动点}$$

$$n=1: y_1(x) - y_0 = -AF(x, y_0)$$

$$\text{取 } F(x_0, y_0) = 0, \quad F(x, y) \text{ 连续}$$

$$\therefore x \rightarrow x_0 \text{ 时: } y_1(x) - y_0 = -AF(x, y_0) \rightarrow 0$$

$$\exists \eta \in (0, \eta_1]: \Rightarrow |AF(x, y_0)| \leq \frac{1}{2} b_1;$$

$$\Rightarrow |y_1(x) - y_0| = |AF(x, y_0)| \leq \frac{1}{2} b_1 \leq b_1;$$

$$\text{由 } \frac{1}{2} b_1 \leq b_1 \text{ 内证: } |y_n(x) - y_0| \leq b_1, \quad \forall |x - x_0| \leq \eta, \quad \forall n \geq 1.$$

假设对 n 成立:

$$\text{对 } n+1: y_{n+1}(x) - y_0 = y_n(x) - AF(x, y_n(x)) - y_0$$

$$= (y_n(x) - y_0) - A(F(x, y_n(x)) - F(x, y_0))$$

$$- AF(x, y_0). \quad (x, \zeta_n(x)) \in D_1$$

中值定理

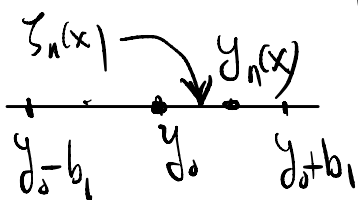
$$= (y_n(x) - y_0) [1 - AF_y(x, \zeta_n(x))]$$

$$- AF(x, y_0)$$

$$\leq \frac{1}{2}$$

$\zeta_n(x) \in [y_0 - b_1, y_0 + b_1]$

与 $y_n(x)$ 之间



$$|y_{n+1}(x) - y_0| \leq \frac{1}{2} |y_n(x) - y_0| + |AF(x, y_0)|$$

$$\leq \frac{1}{2} |y_n(x) - y_0| + |AF(x, y_0)|$$

$$\leq \frac{1}{2} b_1 + \frac{1}{2} b_1 = b_1$$

$$|x - x_0| \leq \eta$$

$$\therefore \forall n \geq 1: \forall |x-x_0| \leq \eta, |y_{n+1}(x) - y_0| \leq b_1.$$

$$\therefore \forall n \geq 1: \forall |x-x_0| \leq \eta:$$

$$|y_{n+1}(x) - y_n(x)| \leq \frac{1}{2} |y_n(x) - y_{n-1}(x)|$$

$$\leq \frac{1}{2} \cdot \frac{1}{2} |y_{n-1}(x) - y_{n-2}(x)|$$

$$\leq \dots \leq \frac{1}{2} \cdot \frac{1}{2} \cdot \dots \cdot \frac{1}{2} |y_1(x) - y_0|$$

$$= \frac{1}{2^n} |y_1(x) - y_0| = \frac{1}{2^n} |AF(x, y_0)| \quad M = \max |AF(x, y_0)|$$

$$= \frac{1}{2^n} M:$$

$$\Rightarrow \{y_n(x)\} \text{ satisfies Cauchy's condition: } \forall \epsilon > 0, \exists N \text{ such that } |y_n(x) - y_m(x)| < \epsilon \text{ for } n, m > N.$$

$$\forall n > m \geq 1:$$

$$|y_n(x) - y_m(x)| \leq |y_n(x) - y_{n-1}(x)| + |y_{n-1}(x) - y_{n-2}(x)| + \dots + |y_{m+1}(x) - y_m(x)|$$

$$\leq \frac{1}{2^{n-1}} M + \frac{1}{2^{n-2}} M + \dots + \frac{1}{2^m} M$$

$$\leq \left(\frac{1}{2^m} + \frac{1}{2^{m+1}} + \dots + \frac{1}{2^{n-1}} + \dots \right) M$$

$$= \frac{1}{2^{m-1}} M \xrightarrow{n > m \rightarrow \infty} 0$$

$$\therefore \{y_n(x)\} \text{ satisfies Cauchy's condition.}$$

Step 3. $y_n(x) \Rightarrow f(x)$: ~~一致收敛~~ $\frac{1}{n} \rightarrow 0$.

① $\{y_n(x)\}$ $\frac{1}{n}$ -Cauchy $\Rightarrow$ $y_n(x) \in C(|x-x_0| \leq \eta)$.

$\therefore \exists f(x)$, $y_n(x) \Rightarrow f(x)$. $\forall |x-x_0| \leq \eta - \epsilon$ ✓

$f(x) \in C(|x-x_0| \leq \eta)$.

~~一致收敛~~ $\frac{1}{n} \rightarrow 0$: $(\exists x)$

Step 4. ~~一致收敛~~ $y=f(x) \in C^1(|x-x_0| \leq \eta)$.

$x, x+\Delta x \in [x_0-\eta, x_0+\eta]$:

$\downarrow$ $\downarrow$
 $f(x)$ $f(x+\Delta x)$ $(x, f(x))$, $(x+\Delta x, f(x+\Delta x))$ 在曲线上;
 $= f(x) + \Delta y$

$$F(x, f(x)) = 0$$

$$F(x+\Delta x, f(x+\Delta x)) = 0$$

$$F(x+\Delta x, y+\Delta y) = 0$$

$$\Delta y \triangleq f(x+\Delta x) - \textcircled{f(x)} = y$$

$$f(x+\Delta x) = y + \Delta y$$

~~恒成立~~: $0 = F(x+\Delta x, y+\Delta y) - F(x, y)$

$$= F(x+\Delta x, y+\Delta y) - F(x, y+\Delta y)$$

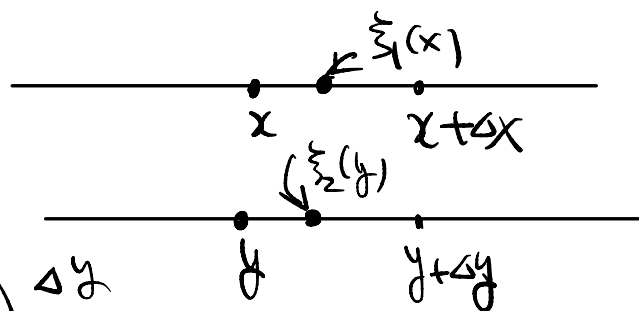
$$F(x, y) \in C^1(D)$$

$$+ \underbrace{F(x, y+\Delta y) - F(x, y)}$$

$$= F_x(\xi_1, y+\Delta y) \Delta x + F_y(x, \xi_2(y)) \Delta y$$

$\xi_1(x)$: 介于 x 与 $x+\Delta x$ 之间,

$\xi_2(y)$ 介于 y 与 $y+\Delta y$ 之间



$$0 = F_x(\xi_1(x), y+\Delta y) + F_y(x, \xi_2(y)) \frac{\Delta y}{\Delta x}.$$

$$\frac{\Delta y}{\Delta x} = - \frac{F_x(\xi_1(x), y+\Delta y)}{F_y(x, \xi_2(y))}.$$

$f(x) \in C[x_0-\eta, x_0+\eta]$:

$$\triangleq \Delta x \rightarrow 0: \Delta y = f(x+\Delta x) - f(x) \rightarrow 0 \quad \begin{cases} \xi_1(x) \rightarrow x \\ \xi_2(y) \rightarrow y \end{cases}$$

$$\frac{\Delta y}{\Delta x} = - \frac{F_x(\xi_1(x), y+\Delta y)}{F_y(x, \xi_2(y))} \xrightarrow[\Delta y \rightarrow 0]{\Delta x \rightarrow 0} - \frac{F_x(x, y)}{F_y(x, y)} \quad \checkmark$$

($\because F_x(x, y), F_y(x, y)$ 连续, $F_y(x, y) \neq 0$)

$$\therefore f(x) \text{ 可导且 } f'(x) = - \frac{F_x(x, y)}{F_y(x, y)}. \quad \#$$

注: ① $F_y(x, y) \neq 0$: 是个充分条件,

例 $F(x, y) = x^3 - y^3 = 0 \Leftrightarrow \exists !$ 隐函数 $y = x$

$$\text{但: } F_y(0, 0) = -3y^2|_{(0,0)} = 0$$

隐函数定理中条件不成立

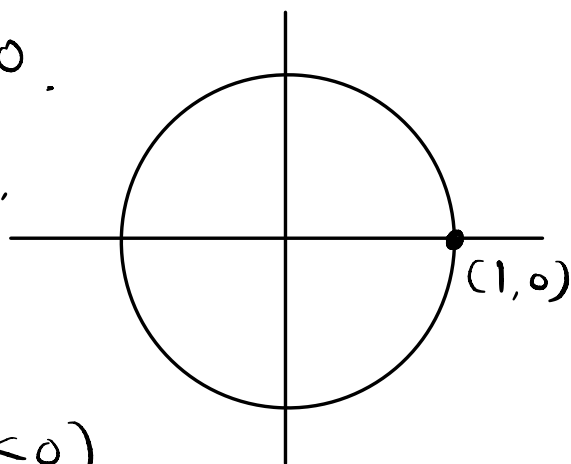
② $F_y(x_0, y_0) \neq 0$ 不满足: 定理不成立:

例 $F(x, y) = x^2 + y^2 - 1 = 0$.

在 $(1, 0)$: $F_y = 2y|_{(1,0)} = 0$.

隐函数 $y = \sqrt{1-x^2} (>0)$

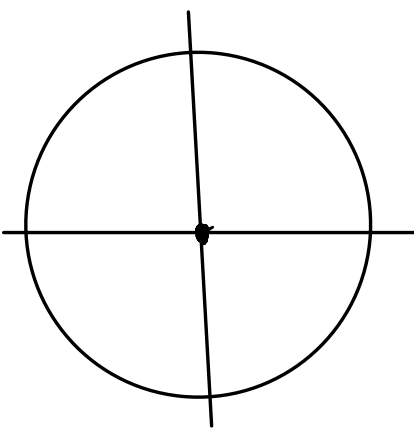
或 $y = -\sqrt{1-x^2} (<0)$.



二. 多元隐函数情形:

$F(x, y) = 0, \quad x \in B(x_0, a):$

$F(x_1, x_2, \dots, x_n, y) = 0 \quad x = (x_1, x_2, \dots, x_n),$
 $|x - x_0| \leq a$



定理 1.2. 设 $F(x, y)$ 定义于 $(x, y) \in D \triangleq B(x_0, a) \times [y_0 - b, y_0 + b]$

$B(x_0, a) \subset \mathbb{R}^n$. 满足:

(1) $F(x, y) \in C^1(D)$

(2) $F(x_0, y_0) = 0,$

(3) $F_y(x_0, y_0) \neq 0.$

则 $\exists \eta > 0, F(x, y) = 0$ 决定了唯一的隐函数 $y = f(x)$. 满足:

(i) $F(x, f(x)) \equiv 0, \quad x \in B(x_0, \eta).$

ii) $y = f(x) \in C^1(B(x_0, \eta))$;

$$\frac{\partial}{\partial x_i} F(x, f(x)) = 0: F_{x_i} + F_y \cdot f_{x_i} = 0$$

$$\Rightarrow f_{x_i} = -\frac{F_{x_i}}{F_y}$$

三. 方程组情况:

$$\begin{cases} F_1(x, y, z) = 0 \\ F_2(x, y, z) = 0 \end{cases} \quad \forall x: \exists \begin{cases} y = y(x) \\ z = z(x) \end{cases}$$

$$\begin{cases} F_1(x, y(x), z(x)) = 0 \\ F_2(x, y(x), z(x)) = 0 \end{cases} \quad \text{or} \quad \begin{cases} y = y(x) \\ z = z(x) \end{cases} : \text{隐函数组}$$

定理 1.3. 设 $F_1(x, y, z), F_2(x, y, z)$ 满足

$$D = \left\{ (x, y, z) \mid \begin{array}{l} |x - x_0| \leq a \\ |y - y_0| \leq b \\ |z - z_0| \leq c \end{array} \right\} \quad \text{满足:}$$

(1) $F_1, F_2 \in C^1(D)$.

(2) $F_1(x_0, y_0, z_0) = 0, F_2(x_0, y_0, z_0) = 0$.

(3) $\frac{\partial(F_1, F_2)}{\partial(y, z)} \Big|_{(x_0, y_0, z_0)} = \begin{pmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{pmatrix} \Big|_{(x_0, y_0, z_0)} \neq 0$ 雅可比行列式非零.

$\text{由 } \exists \eta > 0, \exists \begin{cases} F_1(x, y, z) = 0 \\ F_2(x, y, z) = 0 \end{cases} \quad \text{决定了 "唯一-隐函数定理"}$

$\begin{cases} y = y(x) \\ z = z(x) \end{cases} \in C^1(|x - x_0| \leq \eta), \quad \text{满足:}$

$\begin{cases} F_1(x, y(x), z(x)) \equiv 0, \\ F_2(x, y(x), z(x)) \equiv 0, \end{cases} \quad x \in: \{x \mid |x - x_0| \leq \eta\}.$

$\text{且 } \begin{cases} y_0 = y(x_0), \\ z_0 = z(x_0), \end{cases} \quad y_x = \frac{1}{J} \frac{\partial(F_1, F_2)}{\partial(z, x)}, \quad z_x = \frac{1}{J} \frac{\partial(F_1, F_2)}{\partial(x, y)}$

$\begin{cases} F_{1x} + F_{1y} \cdot y_x + F_{1z} \cdot z_x = 0 \\ F_{2x} + F_{2y} \cdot y_x + F_{2z} \cdot z_x = 0 \end{cases}$

$\text{系数行列式 } J = \frac{\partial(F_1, F_2)}{\partial(y, z)} \neq 0$

证明思路:

$\vec{F}(x, y, z) \triangleq \begin{pmatrix} F_1(x, y, z) \\ F_2(x, y, z) \end{pmatrix} = 0$

$\text{求隐函数且 } \begin{pmatrix} y \\ z \end{pmatrix} = \begin{pmatrix} y(x) \\ z(x) \end{pmatrix}.$

取 A : 非负实数

$$T \begin{pmatrix} y(x) \\ z(x) \end{pmatrix} = \begin{pmatrix} y(x) \\ z(x) \end{pmatrix} - A \vec{F}(x, y(x), z(x))$$

$$= \begin{pmatrix} y(x) \\ z(x) \end{pmatrix} \iff \begin{pmatrix} y(x) \\ z(x) \end{pmatrix}: p \leq z \text{ 收敛}.$$

迭代格式: $\begin{pmatrix} y_0(x) \\ z_0(x) \end{pmatrix} = \begin{pmatrix} y_0 \\ z_0 \end{pmatrix};$

$$\begin{pmatrix} y_1(x) \\ z_1(x) \end{pmatrix} = T \begin{pmatrix} y_0(x) \\ z_0(x) \end{pmatrix} = \begin{pmatrix} y_0 \\ z_0 \end{pmatrix} - A \vec{F}(x, y_0, z_0);$$

$$\begin{pmatrix} y_{n+1}(x) \\ z_{n+1}(x) \end{pmatrix} = T \begin{pmatrix} y_n(x) \\ z_n(x) \end{pmatrix} = \begin{pmatrix} y_n(x) \\ z_n(x) \end{pmatrix} - A \vec{F}(x, y_n(x), z_n(x)),$$

$$J = \begin{pmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{pmatrix} \bigg|_{(x_0, y, z)} \quad \overline{y} \nabla,$$

$$A = J^{-1}: \quad \left| 1 - A \frac{\partial(F_1, F_2)}{\partial(y, z)} \right| \leq \frac{1}{2}$$

记 $Y_n(x) = \begin{pmatrix} y_n(x) \\ z_n(x) \end{pmatrix}$, 则迭代格式为:

$$Y_{n+1}(x) = T Y_n(x) = Y_n(x) - A \begin{pmatrix} F_1(x, y_n(x), z_n(x)) \\ F_2(x, y_n(x), z_n(x)) \end{pmatrix}$$

$$\begin{aligned}
Y_{n+1}(x) - Y_n(x) &= Y_n(x) - A \begin{pmatrix} F_1(x, y_n(x), z_n(x)) \\ F_2(x, y_n(x), z_n(x)) \end{pmatrix} \\
&\quad - (Y_{n-1}(x) - A \begin{pmatrix} F_1(x, y_{n-1}(x), z_{n-1}(x)) \\ F_2(x, y_{n-1}(x), z_{n-1}(x)) \end{pmatrix}) \\
&= (Y_n(x) - Y_{n-1}(x)) - A \begin{pmatrix} F_1(x, y_n(x), z_n(x)) - F_1(x, y_{n-1}(x), z_{n-1}(x)) \\ F_2(x, y_n(x), z_n(x)) - F_2(x, y_{n-1}(x), z_{n-1}(x)) \end{pmatrix} \\
&= (Y_n(x) - Y_{n-1}(x)) - A \begin{pmatrix} F_{1y}(y_n - y_{n-1}) + F_{1z}(z_n - z_{n-1}) \\ F_{2y}(y_n - y_{n-1}) + F_{2z}(z_n - z_{n-1}) \end{pmatrix} \\
&= (Y_n(x) - Y_{n-1}(x)) - A \begin{pmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{pmatrix} \begin{pmatrix} y_n - y_{n-1} \\ z_n - z_{n-1} \end{pmatrix} \\
&= \left[I - A \begin{pmatrix} F_{1y} & F_{1z} \\ F_{2y} & F_{2z} \end{pmatrix} \right] (Y_n(x) - Y_{n-1}(x))
\end{aligned}$$

作註. P 159. 1. 3.